STORAGE



Near-Line SSD: the Path to the Future

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Where does Near-Line fit in the Future?

- Workload considerations: where Near-Line makes sense
- Form-factor: E2 for maximum benefit
  - Why E2? (SDC '25)
- Failure-in-Place: custom fit to the application



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Where Near-Line makes sense

Workload attribute	Capacity I/O	Capacity Cloud Storage Optimized
I/O block size Block size matters	4 KiB - 16 KiB random, sequential *stream*	128 KiB – 256 KiB random, append *streams*
Target write amp	Designed for higher 4.0 or above FDP can help reduce WA	Designed for lower 1.5 – 2.0 or below Designed for low WA with FDP
IU size	Small, ~I/O block size 16 KiB	Big, ~I/O block size 64 KiB to 128 KiB
FTL / L2P caching	Doesn't make sense Big small-block RR performance hit	May make sense Less need with large IU Small-block RR not a key use case
Patterns Distribution matters	Designed to 4 corners Broadly targeted	Optimize for log append, big block random and lower WA (FDP)

E2 for maximum benefit



- E2 requirements found in the OCP DC NVMe spec
  - Section 10.9 – E2 Form Factor Requirements
- Why E2 exists
  - Bit density, cost, TCO
  - Maximum NAND, maximum capacity
- Non-NAND BOM cost is cost overhead
  - E2 is optimal in this metric

Near-Line Tailored Failure in Place

- As QLC drives get larger, FIP can help systems manage devices more efficiently
  - Can reduce or eliminate maintenance
- Current method – in situ
  - Drive hides block / plane / die failures by reducing effective OP
  - Drive maintains capacity but loses performance over time as OP drops
  - Drive replacement based on OP loss – judgment call
- FIP
  - Drive shows block / plane / die failures by reducing effective capacity
  - Drive maintains performance with controlled capacity loss (easier to manage in the fleet)
  - Drive replacement based on effective capacity per slot – value metric

Simple FIP Example

- 1,024TB SSD, 4Tb die, 10% performance OP, 0.1% die AFR, 10Y oplife
- 1,000 SSD cluster, target 10% prob exceeding spares/cluster/oplife
- In situ spares
  - 42 spares/drive (1.9%) = 1.4e-4 probability of exceeding spare count
  - 13% probability of an SSD exceeding spare count in the cluster
- Cluster-level spares (FIP)
  - 1% cluster spares = 13% probability of exceeding cluster spare count
- FIP increases capacity by 0.9% vs. in situ here (~1/2 the spares = 10PB savings)
  - Expected, since global spares will improve the efficiency

FIP and OCP

- OCP is the ideal place to drive the superior approach

- Presuming FIP is technically and economically sound, it still has a fundamental weakness
  - Reporting failed die/blocks looks like a reliability loss
  - Vendors may want to hide this (and go back to OP) to game the metric

- OCP could craft a system-level spec for loss rate(s)
  - If systems designed to spec, no extra credit for artificially exceeding it

Call to Action

- Deploy Near-Line for the workloads that make sense for it
- Start designing for High-Capacity Storage
  - E2 increasingly worth a serious look
  - E3.L also a solid option
- Optimize failure-in-place to maximize near-line benefits
  - Move forward on new standard direction for FIP behavior and reporting
  - Identify best practices – avoid vendor gaming and specmanship

Thank You!

